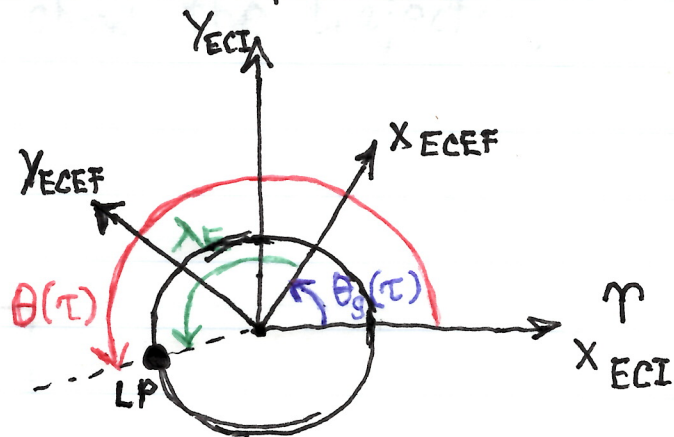


⇒ Rocket Traj. Sim. Step 15



$\theta_g(\tau)$ = Greenwich sidereal time

λ_E = observer's east longitude

$\theta(\tau)$ = local sidereal time

LP = launch point

τ = Vernal equinox

- Sidereal time is the time measured relative to the stars, defined as the angle betw. the observer & the vernal equinox

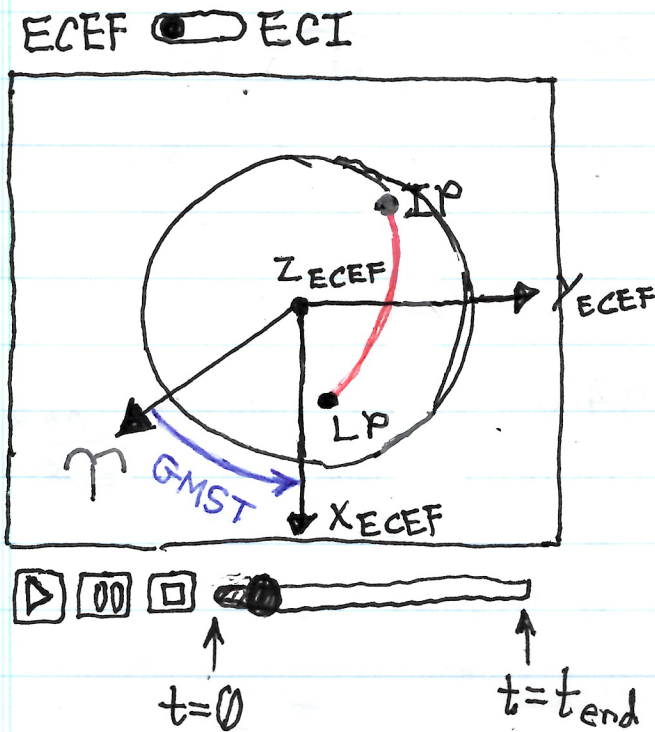
• Greenwich Apparent Sidereal Time (GAST):
Greenwich Sidereal Time which accounts for Earth's nutation & precession

- Greenwich Sidereal Time:
angle measured from vernal equinox to Greenwich meridian.

- Greenwich Mean Sidereal Time (GMST):
Greenwich Sidereal Time which doesn't account for nutation & precession of Earth

~~Converting ECEF to ECI~~

- 3D view should show the trajectory in ECEF frame



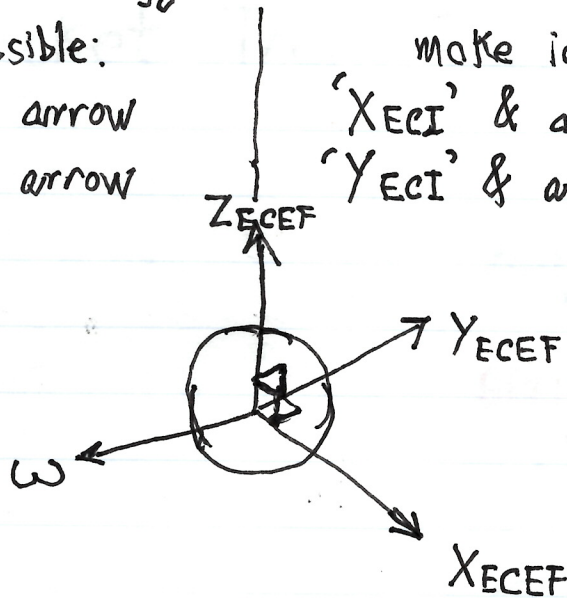
Υ = Vernal equinox

⇒ Upon toggle $ECI \rightarrow ECEF$:

make visible:
' X_{ECEF} ' & arrow
' Y_{ECEF} ' & arrow

make invisible:
' X_{ECI} ' & arrow
' Y_{ECI} ' & arrow

mutate:
' $Z_{ECI} \rightarrow Z_{ECEF}$ '
' $X_{ECI}, \omega \rightarrow \omega$ '



⇒ Upon toggle $ECEF \rightarrow ECI$:

make visible:
' Y_{ECI} ' & arrow

make invisible:
' X_{ECEF} ' & arrow
' Y_{ECEF} ' & arrow

mutate:
' $Z_{ECEF} \rightarrow Z_{ECI}$ '
' $\omega \rightarrow X_{ECI}, \omega$ '

